

Aya AbdElnasser *Junior Data Engineer*

✉ aya.dataengineer@gmail.com ☎ 20-112-426-1378 📍 Naser City Cairo, Egypt 🔗 [linkedin.com/in/aya-abd-el-nasser](https://www.linkedin.com/in/aya-abd-el-nasser)

🐙 AyaAbdElnasser ayadigiliansDE

PROFESSIONAL SUMMARY

Junior Data Engineer with hands-on experience designing and implementing end-to-end data pipelines using Apache Spark, Kafka, Airflow, dbt, Docker, SQL Server, and BigQuery. Skilled in ETL/ELT development, data warehousing, batch and streaming data processing, and workflow orchestration. Built production-style projects involving real-time data ingestion, dimensional modeling, data quality validation, observability, and analytics dashboards. Passionate about building scalable and reliable data platforms.

EXPERIENCE

Data Engineer Trainee

Digilians – Digital Pioneers Initiative

Jan 2026 – Jul 2026 | Cairo, Egypt

- Completed an intensive Data Engineering training program.
- Built batch and streaming data pipelines using Python and SQL.
- Applied ETL, dimensional modeling, and data quality techniques.
- Developed workflows using Spark, Airflow, Docker, and SQL Server.
- Collaborated with teams using Git and GitHub.

EDUCATION

Digital Pioneers Initiative (MCIT) | Data Engineering Track | 2026

Bachelor of Arts and Education Ain Shams University

CERTIFICATIONS

Microsoft Certified: Azure Data Fundamentals (DP-900)

Microsoft Certified: Fabric Data Engineer Associate (DP-700)

PROJECTS

Pharmacy Stock Monitoring AI Data Platform

Tech Stack: Python, Kafka, Spark Structured Streaming, Airflow, dbt, BigQuery, Docker, MLflow, Streamlit.

- Designed and implemented an end-to-end pharmacy stock monitoring platform with real-time data ingestion and analytics.
- Developed Kafka producers to simulate pharmacy sales, inventory, expiry, and return events with schema validation using Pydantic.
- Built Spark Structured Streaming pipelines to consume Kafka topics, process streaming events, detect inventory anomalies, and load curated data into BigQuery.
- Developed dbt transformation models to create analytics-ready marts for daily sales, inventory KPIs, and returns analysis.
- Designed Airflow DAGs to orchestrate data quality validation, mart generation, ML retraining, observability monitoring, alert generation, and vector index updates.
- Implemented automated operational alerts for low stock, expiry risk, and abnormal product returns.
- Containerized the complete platform using Docker Compose, integrating Kafka, Spark, PostgreSQL, Airflow, MLflow, and Streamlit.
- Implemented monitoring and observability workflows to validate data availability, warehouse outputs, ML artifacts, and RAG resources.

<https://github.com/ayadigiliansDE/pharmstock-ai-data-platform.git> 

End-to-End Sales Data Warehouse | ETL Pipeline

- Designed and implemented an end-to-end ETL pipeline using **SQL Server Integration Services (SSIS)** to populate a **Sales Data Warehouse** from OLTP source systems.
- Built a **Star Schema** dimensional model consisting of **FactSales** and multiple dimension tables to support analytical reporting.
- Implemented **Slowly Changing Dimension (SCD Type 2)** for **DimCustomer** to preserve historical customer records.
- Developed SSIS Data Flow packages utilizing **Lookup Transformations** to validate business keys and maintain referential integrity during fact loading.
- Orchestrated ETL workflows through SSIS **Control Flow** to automate dimension loading prior to fact table processing.
- Ensured data consistency and integrity across customer, product, salesman, and sales fact tables within Microsoft SQL Server.

Tech Stack: SQL Server Integration Services (SSIS), Microsoft SQL Server, T-SQL, ETL, Data Warehousing, Star Schema, SCD Type 2

<https://github.com/ayadigiliansDE/End-to-End-Sales-DataWarehouse-ETL.git> 

SKILLS

Programming: Python, SQL

Data Engineering: ETL, ELT, Data Warehousing, Data Modeling, Star Schema

Big Data & Streaming: Apache Spark (PySpark), Apache Kafka

Workflow & Transformation: Apache Airflow, dbt

Databases: SQL Server, PostgreSQL, BigQuery

Cloud: Google Cloud Platform (BigQuery)

Tools: Docker, Git, GitHub, Pandas, NumPy, Streamlit, MLflow, ChromaDB, Terraform, Pydantic

Soft Skills

• Problem Solving • Analytical Thinking • Communication • Teamwork • Continuous Learning • Documentation